MIL-G-20473B (MU)
7 August 1967
SUPERSEDING
MIL-G-20473A
25 July 1956

MILITARY SPECIFICATION

GRENADE, RIFLE, SMOKE, COLORED, M22A2 LOADING, ASSEMBLING, MARKING, AND PACKING

1. SCOPE

1.1 This specification covers the loading of the stabilizer assembly and the assembling, marking and packing of the components of one type of rifle smoke grenade designated as the M22A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-F-202 — Felt, Hair

C-F-206 — Felt, Sheet, Wool, Pressed

UU-P-31 — Paper, General Specification and Method of Testing

MILITARY

MIL-A-2550 — Ammunition and Special Weapons; General Specification for

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC - STD - 105).

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-1167 — Ammunition Data Cards

MIL-STD-1168 — Lot Numbering of Ammunition

MIL-STD-1235 — Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1330

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

82–0–117 — Grenade, Rifle, Smoke, Colored M22A2 Assembly and Marking Diagram

76–1–1349 — Ammunition Container, Metal, M-234A1, for Grenade, Rifle, M-22A2, Assembly, Details, Packing and Marking

76–1–1350 — Box, Packing, Ammunition for Grenade, Rifle, M22A2, Assembly, Details and Packing

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

IEL – 7153359 — Index of Inspection Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Parts. The parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (dwg.) 82-0-117 and with all requirements specified in applicable specifications.

3.3 Functioning. The rifle grenade shall not fail to function and comply with the following requirements when tested as specified in 4.4.1.

3.3.1 *Premature burst.* There shall be no premature burst (see 6.4).

3.3.2 *Stabilizer tube failures.* The stabilizer tube shall not crack, rupture or show evidence of fin detachment.

3.3.3 *Smoke color.* The rifle grenade shall not emit smoke of a color other than that specified.

3.3.4 *Impact delay.* The delay time from ground impact to smoke emission shall be not more than fifteen (15) seconds maximum (max.).

3.3.5 *Smoke emission time.* The rifle grenade shall emit a continuous cloud of smoke for a time period of not less than thirty (30) seconds *minimum (min.)*.

3.4 Repainting and remarking. The contractor shall repaint and remark all component furnished to him painted and marked when the original painting and marking has been damaged or obliterated.

3.5 Workmanship. All parts shall be loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be complete and sharply defined.

3.6 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

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4.1 General quality assurance provisions. Unless otherwise specified in the contract, purchase order, or this specification, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained by defect code for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Statement of findings on all compositions and powders, and certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First article inspection.

4.2.1 *Submission.* The contractor shall submit a first article sample to an approved laboratory designated by the contracting officer. The first article sample shall consist of thirty (30) assemblies completely loaded and packed as required by this specification, and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and material including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials (see 4.1.1.e). A sample con-

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taining known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspections to be performed.* Loaded assemblies will be subjected by the Government to any or all of the examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings. Functioning tests shall be in accordance with 4.3.3.1 of this specification.

4.2.3 *Rejection.* The first article sample shall be rejected if a critical defect occurs or if two or more samples exhibit any major defect.

4.3 Inspection provisions.

4.3.1 *Lot formation.* The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain:

a. Parts of one lot interfix number from one manufacturer.

b. Smoke charge loaded assemblies of one lot interfix number from one supplier.

c. Primers from not more than one lot interfix number from one manufacturer.

Inspection lots shall comply with Standards MIL–STD–105 and shall be numbered in accordance with Standard MIL–STD–1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.3.2 *Examinations.* Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in Standard MIL–STD–105. Continuous sampling plans in accordance with Standard MIL–STD–1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 *Assembly, prior to assembling loading assembly (see dwg. 82–2–41 covering a detail of dwg. 82–0–117).*

Categories	Defects	Method of Inspection	Code No. (see 6.3)
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Length, max.	Gage	01001
102.	Primer missing or inverted	Visual	01002
Minor:	AQL 1.50 percent		
201.	Detonator retainer seal missing	Visual	01003
202.	Detonator retainer not fully seated	Visual	01004
203.	Evidence of poor workmanship (see 3.5)	Visual	01005

4.3.2.2 *Assembly (see dwg. 82–0–117).*

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Safety band missing, loose or damaged to extent that function may be impaired	Visual-Manual	02001
2.	Smoke color marking incorrect	Visual	02002
Major:	AQL 0.65 percent		
101.	Body plug loose or not seated	Visual-Manual	02003
102.	Loading assembly loose on stabilizer assembly	Manual	02004
103.	Fin damaged	Visual	02005
Minor:	AQL 1.50 percent		
201.	Length, max.	Gage	02006
202.	Evidence of poor workmanship (see 3.5)	Visual	02007
203.	Marking, other than smoke color, incorrect, misleading, or unidentifiable	Visual	02008
204.	Bare spot in protective coating	Visual	02009

4.3.2.3 *Container, prior to sealing (see dwg. 76–1–1349).*

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Signal inverted	Visual	03001
102.	Packing component missing	Visual	03002
Minor: None defined.			

4.3.2.4 *Container, sealed (see dwg. 76–1–1349).*

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Smoke color missing or incorrect	Visual	04001
Major:	AQL 0.40 percent		
101.	Key missing	Visual	04002
102.	Container damaged	Visual	04003
Minor:	AQL 1.50 percent		
201.	Marking, other than smoke color, incorrect, misleading or unidentifiable	Visual	04004
202.	Contents loose	Manual	04005
203.	Evidence of poor workmanship (see 3.5)	Visual	04006

4.3.2.5 *Wood packing box, prior to sealing (see dwg. 1–1350).*

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Item or packing component missing	Visual	05001
102.	Item incorrect or insufficient quantity	Visual	05002
Minor: None defined.			

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4.3.2.6 *Wood packing box (see dwg. 76–1–1350).*

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Box damaged to extent that that contents are exposed or liable to become exposed	Visual	06001
102.	Hardware or strapping missing, broken, or loose	Visual-Manual	06002
Minor:	AQL 1.50 percent		
201.	Contents loose	Manual	06003
202.	Hardware or strapping improperly assembled	Visual-Manual	06004
203.	Car seal missing, unsealed, or improperly positioned	Visual-Manual	06005
204.	Marking incorrect, misleading, or unidentifiable	Visual	06006

4.3.3 *Production testing.* Unless otherwise specified herein, the following tests shall be performed using Standard MIL–STD–105 for sampling plan requirements.

4.3.3.1 *Functioning.* The grenade assemblies shall be observed for any failure to comply with the requirements as classified in Table I when tested as specified in 4.4.1.

TABLE I

Defect	Classification	Code No.
Premature burst (see 3.3.1)	Critical	02010
Stabilizer tube failure (see 3.3.2)	Critical	02011
Incorrect smoke color (see 3.3.3)	Critical	02012
Impact delay over max. (see 3.3.4)	Major	02013
Smoke emission time less than min. (see 3.3.5)	Major	02014
Failure to function (see 3.3)	Major	02015

4.3.3.1.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, eighty (80) smoke grenade assemblies shall be selected for test. The lot shall be rejected if a critical defect occurs or if four (4) or more assemblies exhibit any of the major defects listed in Table I.

4.3.3.1.2 After three (3) consecutive lots have complied with the criteria of 4.3.3.1.1, a single sampling plan in accordance with code letter G and an AQL of 2.50 percent shall be used for each lot. The lot shall be rejected if a critical defect occurs or if the number of major defectives exceed the limit of the sampling plan.

4.3.3.2 Moisture content of packing components (see Table II) Major defects.

TABLE II

Material	Code No.
Hair felt	07001
Pressed felt	07001
Chipboard	07003

One sample of each material of sufficient size to perform the tests specified in 4.4.2 shall be selected from each quantity of material used in packing an item lot. If the moisture content of a material sample exceeds the requirement, the material represented by the sample shall be rejected.

4.3.3.3 *Leakage test (see dwg. 76–1–1349).* Major defect Code No. 04007. This test shall be performed 100 percent in accordance with the procedure specified in 4.4.3. Any

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assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–7153359 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Functioning.* This test shall be performed at a Government proving ground. The grenade assemblies shall be launched from an M16 rifle loaded with an XM195 cartridge with the rifle at an elevation of approximately forty-five (45) degrees angle from the vertical. Projection shall be at maximum range for impact on medium to hard rock free ground. Each grenade assembly shall be observed for compliance with the requirements.

4.4.2 *Moisture content.*

4.4.2.1 *Hair felt.* The moisture content of the hair felt shall be determined in accordance with Federal Specification C–F–202.

4.4.2.2 *Pressed felt.* The moisture content of the pressed felt shall be determined in accordance with Federal Specification C–F–206.

4.4.2.3 *Chipboard.* The moisture content of the chipboard shall be determined in accordance with Federal Specification UU–P–31.

4.4.3 *Leakage test.* The packed sealed container shall be tested in accordance with the procedure and equipment specified in 4.3.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* The signal assemblies shall be preserved and packaged in accordance with dwg. 76–1–1349.

5.2 Packing

5.2.1 *Level A.* The sealed containers shall be packed in accordance with dwg. 76–1–1350.

5.3 Marking. Marking of the containers and packing boxes shall be in accordance with dwgs. 76–1–1349 and 76–1–1350.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

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a. Title, number and date of this specification.

b. Ammunition data cards shall be prepared for each lot in accordance with Standard MIL-STD-1167.

c. Submission of first article quantity. Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out, may be obtained from the contracting officer.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Two copies of the designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL-D-45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army Design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs

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must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the details and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic full size prints of top quality with black lines on white background, and shall provide a legible print produced via the 35MM microfilm process. The contractor shall submit one rolled or flat set copy of the final approved design.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Premature burst. A premature burst is defined as detonation of the grenade at firing or during flight.

Custodian:

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Preparing activity:

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